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Chapter 8: The Creative Aspect of Human Language Use: How Modern Language Models Fare with an Old Idea

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Abstract

Ordinary human language use appears to be uncaused, often novel, yet appropriate to the situations in which it is used. This is operationalized by generative linguists like Chomsky as *stimulus-freedom*, *unboundedness*, and *appropriateness*. Taken together, these characteristics locate human language use outside the bounds of determinacy and randomness. This "creative aspect of language use," or CALU, has downstream implications for explanatory theory that are deeply embedded in the generative linguistic enterprise. These include the invocation of a competence-performance distinction that segments linguistic behavior off from abstract knowledge of language, the internalist orientation of linguistic inquiry, and the postulation of a modular and domain-specific generative language faculty. CALU and its implications have been sorely neglected by recent enthusiasm around the use transformer-based Large Language Models (LLMs) as models or theories of human language, failing to account for the depth of the problem it poses for a science of human language and failing to impose the requisite theoretical constraints. Such approaches are fundamentally deficient and limit the role of LLMs in linguistic inquiry.

"It is not a novel insight that human speech is distinguished by these qualities, though it is an insight that must be recaptured time and time again. With each advance in our understanding of the mechanisms of language, thought, and behavior, comes a tendency to believe that we have found the key to understanding man's apparently unique qualities of mind."

- Noam Chomsky, *Language and Mind.*

Introduction

In March 2023, Steven Piantadosi (2023) published a wide-ranging critique of Noam Chomsky's approach to linguistics.[1] "After decades of privilege and prominence in linguistics," he wrote, "Noam Chomsky's approach to the science of language is experiencing a remarkable downfall" (Piantadosi, 2021, 1). The cause: computational advances embodied in contemporary Large Language Models (LLMs). LLMs, he argued, are "bona fide linguistic *theories*" (Piantadosi, 2023, 7). Their ability to produce human-like grammatical utterances is considered an unprecedented simulation of human language's "high-level properties" (Piantadosi, 2023, 27), made possible without invoking methodological constraints like "competence vs. performance, respect "minimality" or "perfection," and without adherence to the dictum that statistical patterns of unanalyzed data are to be avoided (Piantadosi, 2023, 15).

Optimism for the role of LLMs in linguistic theory—potentially as models of language in their own right—has since reached a fever pitch. This volume comprises some responses to arguments concerning their role and artificial intelligence (AI) generally in linguistics.

This chapter will likewise wade into the debate, though from a different angle. Specifically, it focuses on what Chomsky has traditionally called the "creative aspect of language use," or CALU, an old idea traceable back to Cartesian thought (Descartes, 2009/1637; Cordemoy, 1668). Often conflated with the recursive property of generative grammars (the ability to produce and understand an unbounded range of structured expressions), CALU denotes the full breadth of linguistic performance in Chomsky's work; the *ordinary use* of language, not the abstract knowledge it relies upon.

CALU is typically defined by Chomsky and philosophical commentary on his work as the ability to express thoughts through language over an unbounded range effectively *at will*, with no apparent reliance on local stimuli to engage in such expression, while nonetheless converging with the mental states of others who hear (or otherwise interpret) the remarks. Language use is, in this sense, *appropriate* to situations but not *caused* by situations. It is operationalized according to three criteria: *stimulus-freedom*, *unboundedness*, and *appropriateness to situations* (see, Chomsky, 2009, ch. 1; McGilvray, 2001, 6-13; Asoulin, 2013, 228-232).

[1] The preprint has since been updated and published as a chapter, referenced below.

While there are philosophical arguments concerning the prospects for computational devices (including non-transformer based AI systems) to reproduce CALU (e.g., Carchidi, 2025, Forthcoming), this chapter will instead focus on three points: First, the theoretical constraints entailed by CALU's inclusion in a science of language, namely, the invocation of a *competence-performance* distinction and an *internalist* orientation in the study of language and mind. Second, the cognitive architecture necessary to enable—but not explain—creative language use, namely, a *modular* and *generative language faculty*. Finally, it will make the case in tandem with these points that LLMs, conceived as models or theories of human language, fail to impose the requisite theoretical constraints, fail to offer a viable alternative to account for CALU than generative linguistics, and risk excessive ambition in explanatory theory.

The chapter is structured[2] as follows: it begins with a brief overview of the reception thus far of LLMs in linguistics. Then, it proceeds to Piantadosi's critique of Chomsky in particular, as a useful foil for the overall argument. It then identifies CALU as the missing thread in linguistics debates about LLMs. A full explication of CALU is then provided, followed by an illustration of how LLMs fail to replicate it. From there, implications of the foregoing arguments for linguistic theory are teased out, including the necessity of a competence-performance distinction, the internalist orientation of linguistics and the postulation of a modular and generative language faculty, and the excessive ambition of LLMs conceived as theories of human language.

Large Language Models and Generative Linguistics

Piantadosi's (2023/2024) critique represents the latest episode in a long-running drama about the utility of generative linguistics, particularly as articulated by Chomsky. The piece drew from the momentum generated by the success of OpenAI's (2022) ChatGPT-3.5, to which Chomsky, Roberts, and Watumull (2023) had already responded in *The New York Times* in March 2023.

To be sure, the human-like output of LLMs rightfully triggers a rethink of fundamental claims pertaining to themes in generative linguistics. Indeed, Piantadosi is far from alone in this respect—many scholars see direct implications for the study of human language from the rise of contemporary language models (e.g., Beguš, Dąbkowski, and Rhodes, 2023; Kallens, Kristensen-McLachlan, and Christiansen, 2023; Mahowald et al., 2024).

Perspectives vary, with different theoretical weight assigned to LLMs among sympathetic scholars. For example, Futrell and Mahowald (2025) argue that LLMs and theoretical linguistics are mutually beneficial, aiding the development of one another in various respects. In this way, they take a somewhat less strident position than Ambridge and Blything, who argue that LLMs are "*better* theories than any other that have been proposed" (Ambridge and Blything, 2024, 34).

More sympathetic to the early theoretical work of Chomssky, Portelance and Jasbi (2024) argue that levels of adequacy and discovery procedures identified in those works are relevant to the study of LLMs, noting a difference between the statistical learning methods of earlier n-gram

[2] I appreciate reviewers' feedback on the length of this chapter, which has been pared back.

(Markov) models and LLM learning today.[3] They later extended their framework in service of the claim LLMs and generative linguistics are compatible (Portelance and Jasbi, 2025).

Others draw a sharper line between the two. Volenec and Reiss (2025, 2-3) argue that proponents of LLM-driven theorizing like Piantadosi (2023/2024) mistake the research paradigm's orientation toward an account of *linguistic knowledge* (its content and acquisition) for an account of *acceptability judgments*. The competence-performance distinction, they argue, is crucial to generative linguistics yet neglected in critiques, with a distinction between the object of inquiry (the human language faculty) and sources of evidence (acceptability judgments, eye-tracking data, brain imaging, etc.) resulting from it (Volenec & Reiss, 2025, 3-5).

Piantadosi's Critique of Chomsky's Approach

Piantadosi's critique, given its focus, provides a useful foil for our purposes.

It is worth noting that while principally framed as a direct critique of Chomsky's "approach," Piantadosi's critique frequently veers into the broader territory of generative linguistics. This may be because, as Piantadosi (2024, 385-387) himself acknowledges in the critique's postscript, the nature of *scientific inquiry* seems to be lurking behind arguments between himself and his critics.

The central claims by Piantadosi are (1) LLMs are linguistic theories in their own right; and (2) These theories undermine the Chomskyan approach to linguistics. Piantadosi argues that LLMs, because they "develop representations *of* key structures and dependencies," therefore "should be treated as bona fide linguistic *theories*" (Piantadosi, 2024, 360). He is adopting the position of Baroni (2022) here, who argues that it is "appropriate…to look at deep nets as *linguistic theories*, encoding non-trivial structure priors facilitating language acquisition and processing…a general theory defining a space of possible grammars" prior to language-specific training (Baroni, 2022, 7).[4]

The claimed refutation of Chomsky's approach can be found in the idea that language models capture human-like representations of syntactic and semantic structure—they reproduce the aspects of natural language that linguists seek to explain (Piantadosi, 2024, 362-367). Acknowledging that "any model will necessarily have certain tendencies and biases," Piantadosi thus conceives of "each model or set of modeling assumptions as a possible hypothesis about how the mind may work. Testing how well a model matches humanlike behavior then provides a scientific test of that model's assumptions" (Piantadosi, 2024, 364).

In the postscript, Piantadosi (2024, 385) refers to his argument as a form of *inductive* reasoning, *predicting experimental data*, rather than making deductions *from* experimental data. In this way, the argument is that modern language models shed light on human language because they accurately predict the structure of human language via its simulation. This hints at deeper issues

[3] They also emphasize the potential of neural grammar induction models in this context.
[4] Baroni (2022, 6-7) takes a significantly firmer stance than Piantadosi on the matter of innate priors and language models.

of methodology that will be relevant to generative linguistic methodology and CALU explicated in this chapter.

A Missing Thread

Nevertheless, Piantadosi's critique (and the broader debate) does not touch on the matter of CALU and what that portends linguistic theory.

Surprisingly, nor do most responses to Piantadosi and other scholars sympathetic to LLM-driven linguistic theorizing. (One notable exception is given by Moro et al., who briefly note that LLMs do not exhibit stimulus-freedom and are thus "obviously lacking creativity in this sense" (Moro et al., 2023, 83)).

Perhaps this is not surprising, though. Consider Georges Rey's comment on the matter:

> [Chomsky] discusses this issue so frequently, it would be easy to think it's an essential part of this theory, or even what he takes the data for it to be. As we'll see, however, other than the fact that we utter constantly novel utterances, issues of how we use language, and whether it involves ("free) will" and choice, which may or may not be compatible with "contact mechanics," are, in fact, entirely inessential to his core theory and to the crucial data for it (Rey, 2020, 39).

Chomsky has, on this interpretation, thrown a wrench into linguistic discourse, simultaneously making the significant claim that the human language faculty is freely exercised *without* making this a central—or even essential—part of his theorizing.

But is this picture accurate?

McGilvray notes to the contrary that Chomsky has long considered CALU to represent a set of facts about ordinary human language use with which "any science of language must contend" (McGilvray, 2001, 5). Indeed, Chomsky has sometimes referenced the "Galilean challenge" — the problem of why a "machine may be impelled to act in a certain way, but it cannot be inclined; with human beings, it is often the reverse" (Chomsky, 2017, 1).[5] The establishment of general computability theory in the early-to-mid twentieth century meant that it "became possible, for the first time, to address part of the Galilean challenge directly, even though the earlier history remained unknown" (Chomsky, 2017, 1-2).

Chomsky is likewise clear that certain theoretical constraints central to generative linguistics resulted from this reformulation of the problem of ordinary language use. This includes distinguishing between "the internalized system of knowledge [of language] from the processes that access it" (Chomsky, 2017, 2). That is, between *competence* and *performance*. "The theory of computability enables us to establish the distinction, which is an important one, familiar in other domains" (Chomsky, 2017, 2). The orientation is *internalist* (more on this below).

[5] Rey (2020, 387-388) is, for his part, cool to this notion of behavior that is 'inclined but not impelled.'

Linguistic performance—taken in its full breadth, comprising CALU—presents a set of facts that must be dealt with by an explanatory framework worth its salt; the "why" question of linguistic theory (why it is one way and not another) cannot be properly framed without taking it into account, in some way.

What *is* inessential for Chomsky is an explanation for CALU itself. Chomsky holds that linguistic performance is likely beyond the scope of current and potentially future scientific explanation. His emphasis on addressing "part" of the Galilean challenge meant a reformulation of a problem traceable back to Descartes (1637) and Cordemoy (1668) who held that ordinary language use distinguishes humans (minded creatures) from machines (lacking minds). The rise of computability theory provided the tools to make the reformulation; to conceive of the infinite generativity of a finite system (the brain) and posit an idealized "neurobiological Turing machine" (Watumull and Chomsky, 2020, 4)

Thus, one can, for example, engage with Chomsky's articulations of the Minimalist Program or his thoughts on the evolution of the language faculty without any engagement with this notion of free language use, as none of these more familiar subjects require direct engagement with the problem of uncaused but appropriate language use.

Yet, this does not indicate that CALU is not part of the foundation of Chomsky's linguistic theorizing. The set of facts bound up in CALU have downstream implications for a science of language.

The Creative Aspect of Language Use

A fuller explication of CALU is useful to further flesh out its role in a science of language. While some responses to potential objections to the idea are provided, the focus here is primarily on drawing out its implications for linguistic theory, rather than a comprehensive defense of the idea.[6]

Chomsky (e.g., 2009, 69-77) often cites a phrase attributable to Wilhelm von Humboldt: *infinite use of finite means*. This phrase can be parsed in (at least) two complementary ways. The first emphasizes the ability of human beings to construct, with only finite mental resources, an infinite array of structured linguistic expressions. This is the infinite generativity of language use. The second, compatible with the first, emphasizes the *use* of finite means over an indefinite range of both expressions and circumstances.[7] The phrase helps get at this idea of what Chomsky means by 'free' language use.

[6] For more thorough defenses of the idea, see, Chomsky (1966, ch.1), McGilvray (2001), Asoulin (2013), and Carchidi (2024, 2025).

[7] Popular (and often linguistic) portrayals of this phrase and its association with Chomsky typically emphasize only the first parsing (see, e.g., Summerfield, 2023, 249). However, as Chomsky explicitly notes: "[A] number of professional linguists have repeatedly confused what I refer to here as "the creative aspect of language use" with the recursive property of generative grammars, a very different matter" (Chomsky, 2006, xiv). This is not to say that Chomsky's interpretation of Humboldt's work (being an ancestor of modern generative grammar) is correct, but to say that both parsings are used in his work.

Ordinary language use has the characteristic of being innovative. That is, no individual with a mature and developmentally normal language capacity is confined to a pre-sorted list of sentences. It is *unbounded*.

Additionally, ordinary language use does not appear confined to a pre-defined set of stimuli for which it is recruitable. That is, individuals do not appear strictly reliant on stimuli that one can select in advance to use their language in this innovative fashion. It is *stimulus-free*.

Finally, ordinary language use does not appear to have a pre-specified list of situations for which its use is appropriate to the situation at hand and coherent to others who hear (or otherwise interpret) the remarks. That is, individuals use their language in an unbounded and stimulus-free fashion, though their linguistic expressions converge with the mental states of others. It is *appropriate to circumstance*.

"In a phrase," Collins writes in summary of the idea, "a speaker makes free, infinite use of finite means. It is as if the speaker has harnessed randomness (unpredictable unboundedness) in a meaningful, appropriate manner" (Collins, 2006, 560). Each of the three characteristics of language use described can be explicated independently:

Stimulus-Freedom: Animal communication systems are stimulus-bound in that the expression of mental content appears restricted to circumstances (see, Anderson, 2008, 796-797). Human language use, in contrast, does not appear to occur in some kind of fixed relationship with identifiable stimuli in the local context. It is *detachable* from local, and even *possible* (i.e., fictional), contexts (see, Boeckx, 2010, 36).

Research purporting to find human-like linguistic characteristics in non-human primates does not appear to claim as much for the latter. For example, a recent study claiming that bonobos exhibit "nontrivial" compositionality (the ability to combine elements with independent meanings into new combinations in which these elements modify one another) indeed defines the meaning of a bonobo's signal "as the set of features of circumstances (FoCs) that appear at a rate greater than chance across the signal's occurrences" (Berthet et al., 2025, 104). Such bonobo signals are therefore explicitly tied to the circumstances of their use, at least in this study's framework.

Human language use is not like this. Instead, ordinary language use is *elicited* by local stimuli (we are, say, inclined to respond to remarks directed at us) but not *caused* by them.

Some are chilly to the general idea. Harman (1968, 231-234) argued that both internal and external stimuli *do* control language use. For external stimuli, he argued, language comprehension by a hearer "is, or ought to be, a function of what is said" (Harman, 1968, 232). For internal stimuli, the assertion of stimulus-freedom amounts merely to ignorance of the causal factors underlying linguistic performance (Harman, 1968, 230-232). On this latter point, Rey (2020) levels a similar objection, noting that "the current natural sciences of human bodies seem to indicate that their states and motions are not only "inclined," but every bit as "compelled" as the states and motions of any machine" (Rey, 2020, 387).

On the first point, Chomsky's position on external stimuli traces back to a least his 1959 critique of Skinner's (1957) *Verbal Behavior*. In response to the notion of *stimulus control*, Chomsky says: “We cannot predict verbal behavior in terms of the stimuli in the speaker’s environment, since we do not know what the current stimuli are until he responds” (Chomsky, 1959, 32). That is, identification of a given stimulus to which language use is allegedly fixed (caused by) depends on the individual using language arbitrarily, at which point - and only that point - can the scientist identify a stimulus in the environment.

When Chomsky accuses the behaviorist conception of stimulus control as being merely a guise for “mentalistic psychology” (Chomsky, 1959, 52), he is drawing attention to the intrinsically *post-hoc* effort to explain language use. To try and account for language use by appealing to properties in the physical environment, as Skinner did, “the world *stimulus* has lost all objectivity in this usage. Stimuli are no longer part of the outside physical world; they are driven back into the organism” (Chomsky, 1959, 52). Indeed, there appears to be no “fixed association of utterances to external stimuli or physiological states” (Chomsky, 2009, 60).

Any attempt to provide a serious causal explanation for an individual’s linguistic production will merely be an *interpretation*—and not a scientific one—of many factors about the given context—“[t]his is not the well-defined causality of serious theory…” (McGilvray, 2001, 7; see also, Asoulin, 2013, 230). Individuals may, in the face of a potential stimulus (e.g., being asked a question) say nothing at all.[8]

Internal stimuli seem to pose a different problem. The *brain*, accepted wisdom holds, *causes* language use from within. Although we may not be able to *predict* what an individual may (or may not) say in an arbitrary situation, is this tantamount to suggesting language use is detachable from *any* stimulus, including those internal, as though speaking from a void?[9]

To keep our focus, we note immediately that, *even if* human language use were only free of *external* stimuli, this would still mark a disjuncture from computational models, including LLMs, which output in ways controlled by identifiable external and internal stimuli (see below). But how far can the notion of stimulus-freedom be extended for humans?

A dialogue including, among others, Collins and Rey (see, Collins, 2006, 473-477) on this very point surfaces at least one counter-objection: the *inverse* of the original problem pointed out in Chomsky’s (1959) review of Skinner arises when one tries to locate an internal stimulus. That is, if this assumption is accurate—that language use must have an internal, causal factor owing to some complex interplay of physiological mechanisms—then it is (at least as a reasonable standard of scientific inquiry) incumbent upon the scientist to identify such and such internal event that leads to such and such linguistic expression. Yet, at this point, the scientist is merely pairing up a kind of “putative” cause with a given utterance, rather than offering any systematic explanation (Collins, 2006, 474).

[8] This freedom from stimuli should not be taken to mean that a given stimulus – say, another person’s spoken remark – has no controlling effect on how the receiving individual *perceives* that stimulus. To use a familiar example: a native speaker of English cannot help but hear words spoken in English *as English*. They are *not* free of this perception and, indeed, cannot *choose* to be free of it. They *are* free to say whatever they please in response to spoken English (or any other language) or to say nothing at all.
[9] I appreciate an anonymous reviewer’s feedback in raising this objection, which is useful for clarity.

The problem here, without straying too far from our purpose here, is that the rush to assume that all human behavior can be understood in 'mechanical' terms tries to do away with the *at will* component of human language use; humans speak in ways they *intend* to speak, expressing thoughts they *intend* to express (in ways conformable to situations). Yet, the assumption does not reach the same bar held more clearly in the case of external stimuli: no fixed association of utterances with internal states can be identified.

It also raises the Skinner-esque problem: if my linguistic utterances are generated and spoken through a purely 'mechanical' process, owing to a complex interplay of physiological processes, then why do they converge with your mental state (which is undergoing its own 'mechanical' processes)? In other words: how would stimulus-controlled utterances be meaningfully and appropriately applied to situations?

More than this, attempting to identify a stimulus for an *at will* behavior is nearly a contradiction—the intended expression of thought through language describes a person, not a mechanism. Stimulus-control is simply the wrong concept, inadequate to the task at hand.[10]

Unboundedness: Human language use is novel and innovative owing to its *combinatorics*. This is likely the least controversial characteristic to cognitive scientists. Language use consists in combining and re-combining *finite* parts to form *new meanings* without upper limit on the ability to produce new meanings (Anderson, 2008, 797) (save for non-relevant constraints, like memory, fatigue, etc). The unboundedness of language use thus traces back to language's code—syntax (Moro, 2016, 21-23). The ability to produce infinite novel meanings from finite elements in natural language is sometimes glossed as the *infinite productivity* of language (Huybregts, 2019, 2).[11]

Appropriateness and Coherence to Circumstances: The stimulus-freedom of human language use is revelatory of the third characteristic: rather than being caused by local stimuli, language use is instead connected to external stimuli "by the much more obscure relation of appropriateness" (Chomsky, 1975, 302). Appropriateness is an elusive notion—there is, as McGilvray (2001, 8-18) argues, no theory that is forthcoming that makes it possible to simply list all those conditions in which an utterance is appropriate, as language use is *neither* determined nor random (being stimulus-free and unbounded) *yet* appropriate to others. There is, moreover, no apparent limit on the number of situations in which language use *could* be appropriate.

Individuals nevertheless possess the conceptual resources to judge appropriateness *intuitively* and over an indefinite range of situations. These resources are held to be biologically endowed by the generativists. We may make two certain statements, then, about the "appropriateness" conditions. First, appropriateness cannot refer merely to a *functional* use of language in which it is goal-oriented or otherwise context-dependent *because* it is stimulus-free (McGilvray, 2001, 8-10).

[10] For further defense, see Carchidi (2025).

[11] Explanation notwithstanding, the infinite productivity of human cognition is not an observation exclusive to generativists (see, Piantadosi and Fedorenko, 2017).

Second, language use is bound up in *expressions of thought*.[12] Chomsky thus refers to individuals as judges of appropriateness: It "is recognized as appropriate by other participants in the discourse situation who might have reacted in similar ways and whose thoughts, evoked by this discourse, correspond to those of the speaker" (Chomsky, 1988, 5).

Only *together* do these three components constitute CALU (Baker, 2007, 236-237).

LLMs Do Not Replicate CALU

The natural next question is: do LLMs reproduce CALU?

Let us consider each characteristic:

Stimulus-Controlled: Unlike the problem pointed out in Chomsky's (1959) review of Skinner, a set of stimuli that control the outputs of LLMs can be identified. LLMs' outputs[13] are stimulus-controlled in two primary ways: (1) Given an input, the LLM will generate an output value, unless instructed to the contrary (which reinforces the point); and (2) The LLM will not decide to *not* generate an output value, to *alter* the process of token generation (putting aside the specifics of the operations it performs to transform an input into an output), nor to *initiate* its own conversations *without* instructions to do so.

LLMs' outputs are inextricably tied to the "environment" established by its internal programming and external prompting. While an LLM could be manipulated by human programmers in such a way as to make its stimulus responses variable (e.g., to not respond to every third input), the LLM is ultimately bound to either the internal state imposed by human programmers or the prompt given to the model (or a combination of both).

To be clear, to say that an LLM's outputs are stimulus-controlled is *not* to say that the process it employs to transform an input value into an output value is determined, say, by an external human prompter requesting this or that response. That said, the stimulus-controlled nature of its outputs indicates that the LLM's architecture lacks a kind of operational autonomy; its resources are simply there for the taking, so to speak. As we see below, this is quite unlike the cognitive architecture necessary to enable human stimulus-freedom, namely, a modular and generative language faculty.[14]

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[12] Responses to Chomsky arguing that language is primarily a tool for communication rather than thought, most prominent among recent examples being Fedorenko et al. (2024), wholly neglect Chomsky's characterization of creative language use and what this might mean for its conceptualization as a "tool" (which would assign a function to something that appears not to have pre-defined uses, being stimulus-free). See, Carchidi (2025, Forthcoming) for elaborated responses.

[13] We are speaking about the base model of any conversational system (e.g., ChatGPT), though the point applies equally to applications underpinned by LLMs (see, on this structure, Shanahan, 2023, 4).

[14] This is a *weaker* form of the argument. Colin McGinn (2021), somewhat informally, has argued that the production of linguistic expressions that convey one's thoughts is *necessarily* stimulus-free and, conversely, any organism *lacking* stimulus-freedom cannot engage in the infinite productivity of human language *because* they are bound to stimuli (the expressions being more akin to the reflex of the knee when tapped). I remain agnostic to this argument here, though the reader should at least bear it in mind: it would entail that stimulus-controlled LLMs

LLMs exhibit no independent ability to respond or not respond; they are impelled to act but not inclined.

Weakly Unbounded: LLMs exhibit a weak form of unboundedness, relative to human linguistic competence. They can generate new sentences according to whatever model of a given dataset(s) they acquired during training; according to a particular configuration (see generally, McGilvray, 1999, 85-86). Syntactic novelty has been found experimentally in GPT-2 (McCoy et al., 2023).

However, the unboundedness of CALU is strong in the sense that the meanings of linguistic expressions and their grammatical structures are understood by humans. Human language affords the "intricate regulation of form/meaning pairs," Murphy and Leivada write, and it is this that "constitutes the stuff of syntactic theory, not the organization of strings" (Murphy & Leivada, 2022, 1). Several reasons indicate LLMs do not meet this stronger interpretation, including those detailed in contributions to this volume.

An obvious example is the persistence of "hallucinations" in LLMs – outputting false or wholly fictional outputs authoritatively. The comparison may seem skewed against LLMs, as human beings, after all, both err and lie. But in computational models like LLMs, mistakes cannot be attributed to various performance constraints suffered by humans – there are no comparable distractions that may cause their concentration to break, no fatigue from sleep deprivation that may lead to inaccurate statements, and so on.[15] For humans, these contribute to such errors.

LLMs likewise do not lie, as Moro et al. (2023, 83) note. For humans, lying is not an indication that they cannot properly regulate form/meaning pairs, but an indication that they can in fact use this regulation towards ends unrelated to language. LLMs have no such recourse.

There is also the *kinds* of hallucinations LLMs exhibit. One April 2025 study, for example, noted the discovery of a "generalization bias" in state-of-the-art LLMs tested in clinical trial results summarization tasks. The conclusions of the trials were prone to over-generalization. Humans, as the authors note, are prone to this as well. Yet, amusingly, when nine of ten models[16] tested were explicitly prompted to keep their responses accurate, their resulting outputs were "about twice as likely" to contain generalized conclusions (Peters and Chin-Yee, 2025, 10). Whatever the proper interpretation of this, it does not appear to resemble human behavior.

Other, perhaps less appreciated reasons dovetail nicely with this line of thought. As David Lobina (this volume) details in Chapter 3, adversarial attacks on LLMs—perturbing the input

necessarily lack an ability to meaningfully structure elements into sentences, which would limit their ability to engage in the *unboundedness* of human language.

[15] Interestingly, testing LLMs on the same problems but on different hardware (GPU clusters) caused result instability in one set of experiments testing models' performance stability on mathematics benchmarks (see, Hochlehnert, 2025, 7-8). The results seem to cast doubt on the remarks in the main text. Ironically, the best way to interpret them is with a competence-performance distinction, as the same model performing differently on the same problems across different GPU clusters does appear to be a performance constraint, rather than a lack of competence. But the *kind* of inaccuracy exhibited in these results is inhuman, as any human who otherwise grasps the logic of a set of problems tend not to err the same ways (unless distracted, and so forth). There is also no comparison to be made with humans for the swapping out of underlying "hardware" (a metaphor, at best).

[16] A Claude model was the one exception, on this result.

data of a model in subtle, often bizarre (to humans) ways—effectively ‘disrupts’ the LLM’s comprehension whereas humans tend to remain robust to the changes. This does not indicate that LLMs are regulating form/meaning pairs over an indefinite range but instead serving their intended purpose as function approximators (Chapter 3, this volume).

Perhaps also less appreciated is that computational models’ training procedures are *idealized* relative to human language acquisition, as Dupre (2024) argues. Principally concerned with grammar induction models, Dupre argues that, to be considered adequate models of human learners, models must somehow ‘factor out’ the myriad causal sources beyond language itself (communicative goals, externalization systems, etc.) that comprise a human learner’s dataset.

The point for LLMs is that their usefulness as theories likewise depends on providing an alternative account of multiple sources of data on human language acquisition. LLMs may not clear the bar in this respect, and any resulting competencies attributed to them may mistakenly be conflated with human linguistic competence.

Finally, “reasoning” models—LLMs with significant post-training innovations that lead them to output candidate “chains-of-thought” —throw a curveball. The chains-of-thought resemble the verbalized thought processes of humans. Indeed, the DeepSeek-R1 research paper reports an “aha moment” during DeepSeek-R1-Zero’s training: the researchers described the model as having learned to “allocate more thinking time to a problem by reevaluating its initial approach” (DeepSeek-AI et al., 2025, 8) (the model outputs, when trying to solve an equation, “Wait, wait. Wait. That’s an aha moment I can flag here”) (DeepSeek-AI et al., 2025, 9).

The output is remarkably human-like. The perception that this fuels the acquisition of a general-purpose reasoning capability underlies much of the enthusiasm for these models’ development.

However, such outputs may be misleading. One team of researchers from Arizona State University trained and tested three transformer models on the traces and solutions of an algorithm designed for grid-traversal and path-finding (A* search). Once trained, the models are tasked with constructing plans to navigate a grid and reach an endpoint via legal actions (up, down, left, and right). Bizarrely, the model with the best performance—which showed slightly better out-of-distribution performance, even—was the one trained on A* search chains-of-thought that were deliberately corrupted through random mismatching by the researchers! The implication is that chains-of-thought are not predictive of final outputs (Stechly et al., 2025).

The results indicate that LLMs have not acquired a linguistic capacity that regulates form/meaning pairs, so much as the *model is simply doing something different and unhuman-like*.[17] (Function approximation seems to be the better lens, here.)

Yet, generative linguists do not take their object of study to be the externalization of speech emulated by chains-of-thought. Rather, they study an internal competence—an internalized system of knowledge—distinct from the natural language characters outputted in chains-of-

[17] Although the A* search traces used to train the transformers are not strictly linguistic, the point is fundamental: the prior chain-of-thought does not appear causally responsible for the final output.

thought.[18] That LLM chains-of-thought appear to be causally independent of their final outputs poses only a problem for those who argue that LLMs *are* models of human language and that human language *is* speech or written words. Usage-based approaches specifically have to account for this mismatch. The generative linguist is focused on a different matter, for which LLMs may simply have little to no relation.

LLMs are weakly unbounded, relative to humans.

Functionally Appropriate: LLMs, by virtue of being stimulus-controlled, immediately fail to exhibit the appropriateness criterion. They do exhibit a kind of *functional appropriateness* in the sense that their outputs exist in a functional relationship with their internal programming (any hand-coded instructions provided to it, for example) and external environment (the prompter). Efforts to align the LLM with the expectations of a human end-user indeed results only in the restriction of its outputs to serving goals established in its local environment by the human programmer.[19]

Yet, human language use is *elicited* by local stimuli, not caused by them. LLMs' outputs are not elicited by local stimuli but *caused* by them.

LLMs, in sum, do not exhibit the criteria sufficient to designate their outputs as "creative" use, in our sense.

The Implications of CALU for LLMs and Linguistic Theory

That LLMs do not replicate CALU indicates that they are not *language users* in the sense described here. They do not *use* language in the creative manner operationalized in terms of stimulus-freedom, unboundedness, and appropriateness.

To be sure, debates about LLMs in linguistics tend to set aside this designation of "language user," mirroring CALU's relative absence in much discussion of generative linguistics. Contreras Kallens et al. (2023, 3), for example, while arguing that LLMs demonstrate the significance of statistical learning, explicitly rule out the possibility that LLMs are language users because they lack the requisite cognitive capacities for social interaction. Mahowald et al. (2024, 536) consider the question of language use to be an outstanding question. Piantadosi (2024, 38)

[18] Of course, as Mendívil-Giró notes in Chapter 1 (this volume), LLMs output tokens, not natural language characters (see also, Manova, Chapter 6, this volume).
[19] McGilvray marks a distinction here: creative linguistic productions are stimulus-free, so "being appropriate" is not "being caused by environmental conditions." Nor is it being regularly correlated with the environment" (McGilvray, 2001, 9). He explains that this "is not to say that regular correlation, with or without causation, yields no conception of appropriateness at all" (McGilvray, 2001, 9). Creative appropriateness is not "functional," goal-oriented appropriateness and one cannot capture this criterion merely through "articulate scheme(s) or formula(e)" (McGilvray, 2001, 9). One need not satisfy a standard or speak in reference to an established goal in one's environment for their remarks to be appropriate.

conceives of LLMs as simulations of human language, which appears to rule out their function as language *users*.[20]

The implications of LLMs' lack of creative language use ought to be teased out, beginning with the theoretical constraints necessary to make sense of the cognitive capacity of a human language user.

Competence and Performance

The competence-performance distinction is often raised in the context of human behavior's *variability*; it is not feasible to draw conclusions about the human language faculty based on variable behavior, thus a distinction is drawn between competence (linguistic knowledge) and performance (linguistic behavior). Resulting from this competence-performance distinction is a further distinction between the object of inquiry (the human language faculty) and sources of evidence (acceptability judgments, brain imaging, etc.). Sources of evidence must be *controlled*, and have their variability focused, to properly study the language faculty (see, Reiss and Volenec, 2025, 4-5).

To be sure, this is but one route to the competence-performance distinction. Even among some generative linguists, the thinking normally goes that competence is what is being studied, therefore performance – which is not the object of inquiry – is (temporarily at least) irrelevant. However, this neglects that the set of facts bound up in CALU represent another route to the competence-performance distinction. CALU is both a source of said variability and it raises, apart from other matters of behavioral variability, deeper issues about linguistic inquiry.

We noted above that Chomsky was influenced by work in the formal sciences in the early days of generative linguistics and the establishment of general computability theory in the early-twentieth century. Chomsky and colleagues took stock of Turing's insight that "[i]t is possible to invent a single machine which can be used to compute any computable sequence" (Turing, 1937, p. 241); the infinite generativity from finite means of central importance to generative linguistics.

Indeed, as Mendívil-Giró details, Universal Grammar was formulated to conceive of language as natural, biological object characterized by *recursive enumerability* which is capable of *digital infinity* (Mendívil-Giró, 2018, 861). This computational system recursively produces grammatical sentences over an indefinite range (Mendívil-Giró, 2018, 873-874). This finite object "characterizes an infinite array of 'free expressions,' each a mental structure with a certain form and meaning" (Chomsky, 2007, 45). Universal Grammar is the study of its innate basis (Chomsky, 2007, 45-46). To the point, the term *recursion* was directly inspired by work in mathematical logic, which made the term "*recursive*…equivalent to *computable*" (Mendívil-Giró, 2018, 874).

[20] Lederman and Mahowald (2024) do, to be sure, make the case that the novel text-generation of LLMs does pose a challenge to, among other things, the idea that LLMs are merely cultural technologies for the transmission of existing knowledge. However, this operates too far-afield from our focus to be relevant here.

Yet, this computational system (the language faculty) did not alleviate the earlier Cartesian problem of creative language use in production. Internalized knowledge capable of yielding digital infinity is a separate matter from the externalization of that knowledge, though the distinction makes little sense if one can indeed conceive of infinite generativity from a finite system. The postulation of an inborn computational system meant that it "became possible, for the first time, to address part of the Galilean challenge directly, even though the earlier history remained unknown" (Chomsky, 2017, 1-2).

Part of the challenge. The *use* of this computational system in production is another matter, for which a distinction must be drawn: the abstract knowledge encoded in this system (competence) and the use of this system in production (performance).[21] The level of the individual is still present, so to speak, not resolved by the tools of computability theory (see generally, Carchidi (2025; Forthcoming).

There are more serious issues at play. CALU deals with the uncaused but appropriate expression of thought evidenced through ordinary natural language use. Behavior that is stimulus-free is not *deterministic*, though the utterances – often novel (unboundedness) – are not *random* (appropriate). Being neither determined nor random *yet* appropriate, Asoulin notes that the actual *use* of language represented by CALU is "a scientifically intractable aspect of the world" (Asoulin, 2013, 235).

Thus, one requires an explanatory theory that seeks not to explain the *use* of language in production, but the mechanisms that enable the individual to generate an unbounded array of structured expressions. This approach to inquiry is *internalist* (see generally, Asoulin, 2013). Such a project is only possible if one can conceive of a finite biological system (the human brain) yielding unbounded structured expressions. Computability theory filled this gap, allowing part of the original problem of creative language use to come under the heel of science, hence Universal Grammar's object of inquiry being the innate *basis* of CALU.

CALU, then, is fundamental to a theory of human language but not itself the target explanation. This tracks with McGilvray's observation that CALU, alongside the poverty of the stimulus argument, constitute "the basic observations that lie behind Chomsky's—and other rationalist-romantics'—research strategy or fundamental methodology for the study of language and mind" (McGivlray, 2009, 2). He continues: "[T]his suggests that if you want to construct a science of mind and language, you should avoid trying to construct a science of how people use their minds, and especially their language" (McGilvray, 2009, 2).

It is not, contra otherwise sympathetic scholars like Rey (2020), that CALU is *inessential* to a science of language. It is that the very contours of linguistic theory depend on these observations about use. To neglect them in theory-formation moves the scholar further from the human being they purport to study.

LLMs, as noted, do not reproduce CALU. Insofar as LLMs are simulations of human language (per Piantadosi, 2024) or an "existence proof" of statistical language learning (per Contreras

[21] Internalized knowledge reflects what is called I-language whereas externalized speech is called E-language.

Kallens et al., 2023, 2) or theoretically astute devices (per Ambridge and Blything, 2024), they do not shed any additional light on the stimulus-free, unbounded, yet appropriate use of human language, including any potential explanation of the full problem. As Piantadosi (2024, 367) correctly notes, they make no competence-performance distinction during training which can retroactively be applied to the human's LLM-theory-of-choice.

Yet, if a competence-performance distinction is derived on the basis of facts about ordinary human language use, and LLMs do not reproduce or invoke it, then their ability to serve as models of human language is diminished. The competence-performance distinction is too important for any theory to neglect.

This does not rule out the use of LLMs in linguistics, but it tempers their usefulness.

Stimulus-Freedom and the Modular Language Faculty

Further implications can be drawn, extending the foregoing remarks. One bears on the nature of the language faculty under investigation based on the internalist disposition. A central postulation of early generative linguistics was grounding the idea that humans possess an inborn "faculty" of the human brain/mind (whether localized in the brain was not a determinant) that yields infinite generativity with only finite elements. This postulation was made in part on the basis of applying new intellectual tools to an old problem. Is this postulation justified?

Like the competence-performance distinction, arguing about whether the cognitive architecture underpinning human language is domain-specific or domain-general, and related matters, can proceed in CALU's absence. However, the facts bound up in CALU do have downstream implications here, too, particularly from the *stimulus-freedom* of language use.

Stimulus-freedom in language use indicates that the regulation of form/meaning pairs is being carried out by the mind/brain without reliance on identifiable stimuli; there is no pre-defined set of causes in response to which the language faculty is mechanically set into motion. An individual may simply use language. This is the problem of 'free access' (Collins, 2004, 519-520), the *at will* recruitment and deployment of linguistic resources. (Per the remarks above, the solution to this problem for generative linguists is only partial, characterizing the *enabling mechanism* for this 'free access.')

The cognitive mechanism(s) enabling stimulus-free language use must therefore fulfill two responsibilities: (1) the isolated production of structured linguistic expressions with a discrete set of cognitive resources; and (2) this production must, in some way, interface with other, relevant cognitive systems. A *modular language faculty* capable of unbounded generativity fulfills both. As McGilvray explains:

> The possibility of stimulus-freedom in language use can be seen to result from a modular language faculty. Not just any kind of modularity will do: we need a faculty that utilizes its own resources and with internal prompting produces (in apparent isolation) through its

> own algorithms items in the form of linguistic expressions that are unique to it and yet that "interface" with relevant other internal biological systems (McGilvray, 2001, 12).

Linguistic expressions, often novel in nature, are generated without fixed reliance on external stimuli. A faculty which specializes in the production of structured linguistic expressions must therefore be present to generate such expressions should its resources be recruited. This system, though modular in the sense that it is independently constituted, must also interface with other cognitive systems relevant to the expressions under production (or else the appropriateness of the remark under production would be lacking).

A simplistic input-output system that takes an input value, performs an operation over the input, and deterministically transforms it into an output value would be insufficient to account for this kind of operational autonomy in language use. This is unsurprising given that LLMs, failing to reproduce stimulus-freedom, operate in precisely this manner (the lack of modularity in the LLM architecture as a problem for linguistics has been noted in a more traditional context by Fox and Katzir (2024)).

One could argue that domain-general mechanisms are suited to enabling stimulus-freedom. However, this idea – as though linguistic resources were bounced around the mind mechanically – cannot resolve the matter, as it not only fails to acknowledge the 'free access' of linguistic resources in production but also the isolated nature of linguistic resources this entails. Unless one can find a fixed set of associations for language use, either internal or external, then they cannot be conceived as part of the 'push-and-pull' between cognitive systems. More than this, once broadened to the full scope of ordinary language use, such use must likewise account for the unboundedness and appropriateness of stimulus-free utterances.

This is where the need for this modular language faculty to interface with other cognitive systems arises. The faculty must possess "conceptual resources" that are made "legible" to other systems (McGilvray, 2005, 218). Conceiving of the language faculty as a sub-system of the mind/brain is par for the biolinguistic course (see, e.g., Chomsky, 2013, 35). It is notably a break from the Cartesian approach to the study of mind from which CALU is derived, in which the productivity of human cognition owed to a single source (Asoulin, 2016, 8).

LLMs-as-Theories Are Too Ambitious

Chomsky's inclusion of CALU in scientific theory places limits on the scope of its inquiry. Specifically, the target of explanation ought to be the internal cognitive mechanism (competence) that enables linguistic behavior (performance). Behavior itself—in its entirety—is taken to be beyond the scope of the tools that allow linguists to study a finite computational system. It likewise appears to be *neither* determined *nor* random *yet* appropriate, placing it outside the bounds of current scientific inquiry wholesale.[22] Without such limits, a theory of human

[22] Whether this is a problem in principle or merely owing to current limitations, we put aside here.

language risks being overambitious, perhaps driven by a deficient understanding of the problems at hand.

The seeming rigidity of generative linguistic theory, at least as espoused by Chomsky, is sometimes taken to be part of a pattern in which, when confronted with disconfirming data that threatens his theories, Chomsky clings to the core tenets of Universal Grammar. His invocation of the "Galilean" method (or "style") is often interpreted as such (see, e.g., Behme, 2014).[23]

Interestingly, Piantadosi (2024, 379-380) takes LLMs to be a refutation of the Galilean method.[24] His extended remarks are provided for clarity:

> Often, the only way to study such complex systems is through simulation. We often can't intuit the outcome of an underlying set of rules, but computational tools allow us to simulate and just see what happens. Critically, simulations *test the underlying assumptions and principles* in the model: if we simulate traders and don't see high-level statistics of the stock market, we are sure to have missed some key principles…We don't get a direct test of principles because the systems are too complex. We only get to principles by seeing if the simulations recapitulate the same high-level properties of the system we're interested in. And in fact the *surprisingness* of large language models' behavior illustrates how we don't have good intuitions about language learning systems (Piantadosi, 2024, 380) (emphases in original).

Systems like human language are so complex that the optimal way to test the principles embedded in them is by simulating their use. LLMs perform this simulation, building an internal representation of textual (and other) data based on next-token prediction that gives rise to emergent phenomena. Their success in reproducing human-like language is powerful evidence against the idea that innate, high-level linguistic principles are necessary to give rise to the properties of language in which linguists are interested. From this, Piantadosi believes the Galilean method is refuted.

Piantadosi's remark bears on the relationship between data and theory, decidedly bent towards broad data coverage as a means of explanatory depth. But no real explication is provided of the "method" purportedly employed by Chomsky.

Chomsky's explication of the Galilean method bears on an analogy he makes with CALU. In his 1980 *Rules and Representations*, Chomsky (1980, 8) cites physicist Steven Weinberg's (1976) "The Forces of Nature," which describes the Galilean method in physics as the process of distancing oneself from commonsense ideas about natural phenomena (*abstracting* away from them) and *lowering one's expectations* about what science can realistically accomplish.[25]

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[23] Behme incorrectly attributes the term "Galilean" style to Chomsky, claiming he "coined a term for this unorthodox methodology" (Behme, 2014, 687). The term originates in Husserl (1970/1936, 41).

[24] For this claim, Piantadosi's (2024, 379) only engagement with Chomsky is a 2012 interview between Chomsky and Yarden Katz (see, Katz, 2012). No scholarly works of Chomsky's are referenced to support this claim about the Galilean method.

[25] The emphasis on abstraction and the role of data in the Galilean method is often misunderstood by empiricist-leaning critics. Chater et al. (2015), for example, argue that Chomsky misses two critical principles of Galileo's

Chomsky argues for an analogy between this Galilean style in physics and the method employed by Cartesians: that free human action cannot be meaningfully studied but its enabling mechanisms and influences *are* amenable to scientific inquiry (Chomsky, 1980, 8). He suggests that the successes in physics may be "a remarkable historical accident resulting from chance convergence of biological properties of the human mind with some aspect of the real world" (Chomsky, 1980, 9).

The driving idea behind this analogy is that CALU may *not* have a "chance convergence" of this kind; our science appears inadequate to deal with it. Thus, the intractability of CALU places limits on what a science of language can realistically accomplish. The Galilean method reinforces, by way of analogy, the distinction between competence (enabling mechanisms) and performance (free linguistic behavior).[26]

All this leads to the conclusion that LLMs, by virtue of not postulating any of these theoretical constraints, nor identifying possible theoretical limitations, necessitated by CALU are *too ambitious*. Adopting them as theories, or fundamentally informative for explanatory theory, risks the illusion of progress.

Adopting LLMs as theories of human language fails to grapple with the *kind* of problem that ordinary language use poses and what this entails for a science of human language. A failure to posit a competence-performance distinction, a resistance to internalism and domain-specificity for language's cognitive basis, and so forth, all flow downstream from a theory that puts excessive stock in LLMs. Mahowald et al., for example, note in a sidebar that whether LLMs are "individual language users or…distributions over potential user outputs" is an open question (Mahowald et al., 2024, 536). The uncertainty here is reasonable. However, there is no engagement with the problem of uncaused but appropriate linguistic behavior; this effort to conceptualize LLMs in comparison to humans misses a fundamental piece of the puzzle.

The general push for LLMs-as-theories also risks a retreat into *commonsense* (or what philosophers call *rational*) explanations for linguistic behavior. Excessive reliance on LLMs for explanatory theory-formation may lead one to assume humans use their language in obvious ways, not requiring fundamental constraints on one's inquiry. In our personal lives, we ordinarily appeal to these commonsense explanations for others' behavior ("Jim only said that because he's still mad about what Lucy said the other day"), yet this does not clear the bar for scientific inquiry. As McGilvray (2001, 13-18) aptly explains, this would be like dropping the pretension of "H_2O" and using, instead, our commonsense notion of "water" in chemistry.

The virtue of the Galilean method, and the analogy Chomsky makes between the Cartesian thesis on free human action and modern physics, is its move *away* from commonsense ideas about

theorizing: "first, that we must look not to books but to Nature, the real phenomena, if we are to understand the world, and second, the language in which Nature is written is mathematical, which is to say, quantitative in character" (Chater et al., 2015, 96). This emphasis on "real phenomena" and their mathematical nature mistakes the Galilean method's disposition as one towards data when it is a disposition towards *abstraction* (see, Collins, 2023, 5; Rey, 2020, 20).

[26] For more extended remarks on this, see, Carchidi (2024, 18-20). For a comprehensive explication of the Galilean style in generative linguistics, see, Collins (2023).

human language. It is not obvious what we are dealing with here. It is likewise not obvious how LLMs offer a depth of understanding previously unattainable. The appropriateness of stimulus-free and unbounded human language use—if it is to be explained—cannot have the "rational," everyday account that we provide for it, just as chemists think of water not as the liquid one drinks but as "H_2O." Nor can it be ignored.

None of this rules out the use of LLMs in linguistics as useful tools. Nor should this be viewed as an empty gesture to proponents of their uses as tools of science. Their use, however, must be informed by a theory of human language constructed independently. An immediate implication of this is that, based on the theory one adopts, they will be more useful in some places and less useful in others; they are not theoretical juggernauts, capable of resolving every outstanding problem in linguistics or the broader cognitive sciences. Where LLMs do inform theories, it will likely be second-order in nature; not challenging but instead *adhering to* the fundamental constraints necessitated by the most basic facts about human language.

Conclusion

Perhaps the greatest misconception about Chomsky is that his linguistic work is emblematic of a new scientific method, abruptly departing from the scientific status quo in favor of his own idiosyncratic and ahistorical conception of Universal Grammar. As we have seen, CALU—an old idea—plays a central role in generative linguistic theory, both setting limits on explanatory theory and pointing scholars in directions resulting from those limits. It provides the shape of linguistic theory.

LLMs are likely to continue generating interest in linguistics and the broader cognitive sciences. However, they do not reproduce nor provide insight into CALU previously unavailable to scholars. At best, they are tools to enhance the productivity of linguistic inquiry grounded in theory. At worst, their uses convince scholars that LLMs hard problems like ordinary language use are no bother at all. This chapter has made the case that their uses are limited and any theories built with or without them require those stubborn, old ideas deeply rooted in generative linguistics.

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